1. try-catch

Problem:
Write a program that takes two numbers from the user and divides them. Use `try-catch` to handle:

- Division by zero
- Invalid input (non-numeric)

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2. Multiple catch blocks

Problem:
Write a program to:

- Create a string array of size 2
- Try accessing the 5th element
- Then try to find the length of a null string

Use multiple catch blocks to handle both `ArrayIndexOutOfBoundsException` and `NullPointerException`.

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3. finally block

Problem Statement:
Write a program to simulate **ATM cash withdrawal**.

- Take a hardcoded account balance (e.g., ₹10,000)
- Ask the user to enter an amount to withdraw
- If the amount is greater than balance, throw `ArithmeticException`
- Else, deduct and show the new balance
- **In all cases, the program should print "Transaction session ended." using a `finally` block**, regardless of whether the transaction was successful or failed.

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4. throw keyword

Problem:
Create a method `validateMarks(int marks)`

- If `marks < 0` or `marks > 100`, manually throw `IllegalArgumentException`
- Else print “Valid marks”

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5. Custom Exception

Problem:
Create a custom exception `InvalidPasswordException`.

- Write a method `checkPassword(String pwd)` that throws the exception if the password is less than 8 characters or doesn't contain a digit.
- Handle it in `main()` and print a message.

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6. Nested try-catch

Problem:
Write a nested try-catch program where:

- Inner block handles `ArrayIndexOutOfBoundsException`
- Outer block handles `ArithmeticException`

Trigger both exceptions one after the other.

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7. Exception Chaining

Problem:
Write a method `processPayment()` that:

- Catches a `NullPointerException` and throws a new `RuntimeException` with the original as the cause.

Handle the final exception in `main()` and print the full cause chain.

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8. Full Workflow Mini Task

Scenario: Student Grading System

- Get student name and marks (simulate with hardcoded values)
- Validate:
  - Marks between 0 and 100 — else throw `InvalidMarksException` (custom)
  - Name is not null — handle `NullPointerException`
- Always print “Grading process completed” in `finally`